Converting PFAS Dissociation Constants to Protein–Water Partition Coefficients

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Summary

A PFAS dissociation constant converts to a protein-water partition coefficient exactly, in one line of arithmetic, and the conversion is still not worth much. Six findings, in order of importance.

1. **The conversion is exact but the constant is protein-specific.** `log K = log10(1e9/M) + log n - log Kd`. The leading term is 4.1774 for albumin and 4.8475 for liver FABP. Using albumin's on an L-FABP constant is 4.7x low, silently, with no diagnostic in the output.
2. **Converted values sit 0.86 log below measured ones**, consistently, over the 27 pairs from our best-powered source. That is a missing site count: closing it needs about seven sites where a single-site fit allowed one.
3. **Non-specific binding is not the explanation.** We tested it. At the trace limit the specific term carries 86-98% of the total, so dropping the non-specific term costs 0.07 log at most - twelve times too small.
4. **The input is noisier than any of this.** Two laboratories measuring the same albumin Kd by the same method differ by 1.45 log, 45x on PFOA. A converted log K inherits that one for one, so it cannot be better than 28x reproducible whatever stoichiometry anyone assumes.
5. **Species barely matters here and sex cannot.** Albumin constants move 0.09 log between human and bovine and 0.36 log across four mammals, both below the 0.59 log between-laboratory floor. No binding row resolves sex, because purified albumin has none. The large species and sex effects in PFAS toxicokinetics are in transport, not binding.
6. **Protein does matter, through the slope rather than the level.** Albumin's saturable site is flat in chain length (+0.026/CF2) while globulin, liver FABP and phospholipid membrane all climb at +0.35 to +0.40. Albumin's 1.5 log advantage over globulin at short chains is gone by C10, so an albumin-only plasma model biases the free fraction high for exactly the long-chain compounds that accumulate.

The practical consequence: use a measured partition coefficient where one exists, and treat a converted one as an order-of-magnitude estimate of the specific term alone.

Scope: the ten sources that supply the most measured constants against a named single protein. Four findings rest on papers outside that set; those are named at the end. A later section extends to the whole database and reports what changes.

1. The two quantities

A dissociation constant and a partition coefficient answer different questions about the same experiment, which is why one converts into the other at all.

Kd is the free concentration at which half the sites are occupied. Units of concentration. It describes **one class of site** and says where that site saturates.

K (or D) is the ratio of PFAS in the protein phase to PFAS in water, at equilibrium. Units of volume per mass (or per volume) of protein. It describes **everything the protein holds** and, being a ratio, is a constant only while nothing is saturating.

The same number can be written either way. Three things stop them being interchangeable in practice, and all three occur in this dataset.

Trap	What goes wrong	Size
Basis	per kg of protein vs per litre of protein	0.135 log (1.364x, from albumin's 0.733 mL/g partial specific volume)
Sorbent definition	protein alone vs protein + lipid vs whole serum	unbounded; a whole-matrix coefficient is what a mass balance must *predict*, not an input to one
Speciation	K is one species, D is all species at a stated pH	negligible for the acids; material only for PFOSA (pKa 8.77, 4.1% anionic at pH 7.4)

Fischer 2024 and Allendorf 2019 report per kg; Qin 2024 and Bischel 2011 report per litre. Every number in this document is on the **per-kg** basis, with the density conversion applied where the source used the other one.

2. Derivation

Five steps. The important one is step 3, where the protein concentration cancels.

Step 1 - site occupancy. For one class of site with total protein P and free ligand Cw, the bound complex PL obeys the mass-action definition of Kd:

$$K_d = \frac{[P][C_w]}{[PL]}$$

With the site balance `P_total = [P] + [PL]`, the free protein `[P]` cancels and the fraction of sites occupied is

$$\theta = \frac{C_w}{K_d + C_w}$$

Step 2 - bound amount. If each protein molecule carries n such sites, the ligand bound per litre of the protein phase is

$$C_{bound} = P \cdot n \cdot \theta = \frac{P\,n\,C_w}{K_d + C_w}$$

Step 3 - divide by the protein mass. The partition coefficient is bound ligand per kilogram of protein, divided by the ligand left in water. One litre holding P micromoles of protein of molar mass M weighs `P x M / 1000` grams, so

$$K = \frac{C_{bound}}{C_w} \cdot \frac{1000}{P\,M} = \frac{1000\,n}{M\,(K_d + C_w)}$$

P has cancelled. This is the whole reason the conversion works: a partition coefficient does not depend on how much protein was in the cell, which is exactly what makes it transferable between an assay and a tissue. It does depend on Cw.

Step 4 - the trace limit. As `Cw -> 0`, which is where a partition coefficient is a constant at all:

$$K^0 = \frac{1000\,n}{M\,K_d}$$

Step 5 - put Kd in micromolar and take logs. With Kd in uM rather than molar, the 1000 becomes 1e9:

$$\log K = \log_{10}\left(\frac{10^9}{M}\right) + \log n - \log K_d(\mu M)$$

The `1e9` and not `1e6` is what makes the two sides commensurate. An error here is a silent factor of 1000; in an early draft of this project's code it placed every implied site count between 156 and 117,000.

Interpretation of the leading term. `1e9/M` is the number of micromoles of protein present in one kilogram of that protein - 15,046 for albumin. It is what puts bound amount (mol per mol of protein) and partition coefficient (litres of water per kg of protein) in the same units.

Where this is published. Nothing above is this review's invention; two groups write the same bridge independently, and their equation numbers are worth having to hand.

Chen et al. 2025 (ES&T 59:82) gives it as **eq S4** of its Supporting Information, Eq. 4 in the main text, as D_HSA/W = B_max x rho_HSA / [(K_D + C_aq/MW_PFAS) x MW_HSA]. That is the expression above with B_max standing for n, on a volume rather than a mass basis, and the albumin concentration already cancelled; its MW_PFAS does nothing but convert C_aq from mass units to molar, so it vanishes with the whole term at the trace limit.

Qin et al. 2023 (Environment International) builds it across **Eq. 13 to Eq. 16**: Eq. 13 defines the molar ratio of PFAS to albumin, Eq. 14 is the Langmuir itself, C_bound = C_max x C_w / (K_d + C_w), Eq. 15 is K_specific = C_max / (2 x K_d) at half-saturation, and Eq. 16 adds the non-specific term. Qin also uses the same two constants this document does - 66,463 Da for albumin and a density of 1.36 kg/L - which is why the numbers line up rather than merely the algebra. The companion Qin et al. 2024 paper (ES&T, doi 10.1021/acs.est.3c10824) is where the bovine albumin values used here are tabulated.

3. The leading constant is protein-specific

The value 4.1774 is specific to albumin and is not a universal constant. A protein of lower molar mass packs more binding sites into a kilogram, so the **same Kd gives a larger partition coefficient**.

Masses are the mature chain computed from the UniProt sequence (average residue masses, water added, two hydrogens removed per disulfide), which is what a purified preparation actually weighs.

Protein	M (g/mol)	umol per kg	log10(1e9/M), per kg	per litre
Serum albumin (HSA/BSA)	66,463	15,046	**4.1774**	4.3123
Human L-FABP (P07148)	14,208	70,383	**4.8475**	4.9824

Protein	M (g/mol)	umol per kg	log10(1e9/M), per kg	per litre
Human I-FABP (P12104)	15,106	66,199	4.8209	4.9558
Rat L-FABP (P02692)	14,272	70,067	4.8455	4.9804

Albumin to human L-FABP is **+0.670 log, a factor of 4.68**.

Worked example, lowest-mass protein in the dataset. PFOA on human L-FABP by equilibrium dialysis, single-site fit, Kd = 0.099 uM:

$$\log K = 4.8475 + \log_{10}(1) - \log_{10}(0.099) = 5.85$$

Using albumin's constant by mistake gives **5.18** instead. Both look like reasonable log K values and nothing in the number says which is right.

Species differences inside albumin do not matter. HSA computes to 66,437 and BSA to 66,398 - 0.00025 log apart, three orders below every other uncertainty here. One mass is used for both.

Constants that cannot be converted. Seven PPAR constants have an affinity *and* a stoichiometry and still cannot be converted: the source names only the supplier, never whether the material is the full-length receptor (52.2, 57.6 and 49.9 kDa for alpha, gamma, delta) or the ligand-binding domain alone, about a third of that. A factor of three in M is 0.48 log. The missing input is the **mass of the thing that was weighed**, not the stoichiometry.

4. Two corrections to the trace-limit form

4.1 Several classes of site

Step 2 of the derivation summed over one class. Several classes simply add:

$$K^0 = \frac{10^9}{M} \sum_i \frac{n_i}{K_{d,i}}$$

Worked on the one source in our set that fits PFOA on rat L-FABP twice, by two methods:

Fit	Kd values (uM)	log K0
Displacement, 2 site classes	13.1, 23.8	3.92

Fit	Kd values (uM)	log K0
ITC, 3 site classes	3.1, 26.2, 52.6	4.43

The two differ by 0.51 log (3.2x). The behaviour of the sum is the result of interest: adding a *third* class moved the answer **up**, and not because three beats two - the ITC fit simply found a tighter best site (3.1 against 13.1 uM). That tightest term alone carries **85%** of the ITC sum.

So a sum over sites is robust to missing a weak class and completely exposed to getting the strong one wrong. One laboratory, one preparation, two instruments, still 3.2x apart.

4.2 Saturation regime

The trace limit in step 4 is an approximation. The exact statement, straight from step 3, is

$$K(C_w) = \frac{K^0}{1 + C_w/K_d}$$

At `Cw = Kd` the coefficient is exactly 0.301 log (a factor of 2) below the plateau. Qin reports a specific-binding column on exactly this basis. Qin et al. 2023 (Environment International) defines it in its Eq. 15 as `K_spec = C_max / (2 K_d)`, the ratio **at half-saturation**. Comparing that column with a converted Kd without adding 0.301 compares two different points on the same curve.

For PFOA on L-FABP, Kd = 0.099 uM:

Free Cw	log K
0 (plateau)	5.85
0.099 uM	5.55
1 uM	4.81

A full log unit lost between trace and a typical bioassay concentration.

Knowing which regime a measurement was in is a separate problem from correcting for it. The correction above needs Cw and Kd. What the database can usually supply instead is the ligand-to-protein ratio the assay reached, which asks the same question in the units

these papers actually report: below about 0.01 a measurement sees the specific site, above about 1 it sees non-specific partitioning. 295 of the 1,208 binding rows now carry one, on the measurement rows and on the assay table alike.

One source reports a ratio inconsistent with its own concentrations. Khazaee 2021 states a ligand:protein mole ratio of 0.05 to 5. Its recorded concentrations - 0.33 to 153.5 ng/mL against 1 uM L-FABP - give 0.31 to 1.49 once converted compound by compound. The two differ by seven to seventeen fold and, worse, fall on opposite sides of the line this column exists to draw: 5 is the partitioning regime, 0.31 to 1.49 is the borderline. The workbooks carry the paper's own number with the contradicting arithmetic spelled out beside it, because publishing the computed value would have moved those L-FABP and PPAR measurements out of the partitioning regime on the strength of concentrations that cannot reproduce the paper's own statement.

Bischel 2011 is the benign version of the same shape, and it is why the rule is not simply "believe the paper". Bischel states 0.0034 for every row, and its per-compound concentrations give 0.0013 to 0.0034 - one study-wide ceiling, and then the same experiment described again compound by compound, with both readings firmly in the specific-site regime and nothing at stake between them. An order of magnitude separates that case from Khazaee's. The threshold is written into the code rather than left implicit in whichever number happened to win.

5. The reverse conversion is not identifiable

Algebraically the reverse is trivial - the same equation rearranged:

$$K_d = 10^{\log_{10}(10^9/M) + \log n - \log K}$$

In practice **zero** of the 207 partition-coefficient rows in our set can take it, and that zero is structural rather than a gap in our extraction.

The trace-limit measurement is

$$K^0 = \frac{10^9 n}{M K_d}$$

which determines only the **ratio** `n / Kd`. It cannot separate them, in principle, however well the experiment is done. A study that works in the trace regime - which is what makes its coefficient a constant at all - has thrown away the information needed to recover a Kd.

To get one you need an experiment that reaches saturation, and then you are measuring a Kd directly and do not need the conversion.

The forward and reverse directions are not symmetric, and further data extraction cannot make them so.

6. Converted against measured values

39 pairs as first computed - 29 after the withdrawal described at the end of this document - matched on the canonical compound name (see the naming note below), protein **and** species, never pooled across studies. Every converted value comes from a paper that supplied both the affinity and the stoichiometry; every measured value is Fischer 2024, the only source reporting human albumin partition coefficients on a mass basis.

Converted from	Pairs	Median residual	Where its n came from
Chen 2025	27	**-0.86**	single-site Langmuir, n fixed at 1
Wu 2024	5	-4.01	fluorescence double-log slope
Peng 2024	3	+0.13	fluorescence double-log slope
Chen & Guo 2009	2	-0.53	fluorescence double-log slope
Moro 2022	2	+0.28	ITC, n fitted from the isotherm

Residual = converted minus measured, log units. 32 of 39 are negative.

The one well-powered comparison is 0.80 log low with a consistent sign. 10^0.86 = 7.2, so closing it needs about seven sites where a single-site fit allowed one. Maso 2021 co-crystallised **four PFOA per human serum albumin**, with one high-affinity and three low-affinity sites in solution - independent, structural, and of the right order.

The remaining 12 pairs are too few to read as a trend. They do establish one negative: three papers taking n off a fluorescence double-logarithmic slope land at -4.01, -0.53 and +0.13, so that slope is not a site count. Those ten pairs have since been withdrawn from the comparison for exactly that reason; see the end of this document. Wu 2024's problem is not even its n - its association constants sit three to four orders below every other measurement of the same compound on albumin.

6.1 Non-specific binding as an explanation

The natural story is non-specific binding: a Kd describes one saturable site, a measured coefficient also counts whatever sticks to the rest of the protein, so the converted value should be low by the non-specific share.

Qin 2024 is the only source in our set reporting the two terms **separately**, for bovine albumin, fitted to the same isotherm. At the trace limit:

Compound	nCF	Specific (plateau)	Non-specific	Total	Specific share
PFBA	3	2.61	1.83	2.67	86%
HFPO-DA	4	3.48	2.04	3.49	97%
PFHxA	5	3.43	2.56	3.48	88%
PFHpA	6	4.29	3.33	4.33	90%
PFHxS	6	5.19	3.44	5.19	98%
PFOA	7	4.75	3.75	4.79	91%
PFOS	8	5.44	4.04	5.45	96%

Dropping the non-specific term costs **0.07 log at most**. The gap to explain is 0.86. Non-specific binding is twelve times too small, and the sign of its chain-length trend is wrong too.

Obvious in hindsight: the specific term is saturable and therefore **largest** at trace, which is exactly the limit a converted Kd describes. Non-specific partitioning takes over higher up the isotherm, where neither number applies. The missing capacity is extra **sites**, not extra surface.

7. Magnitude of the residual: five error scales

A residual means nothing without a yardstick. All five of these come from the same database, and the last one changes how the other four read.

What is being compared	Disagreement	Fold	n
One lab, same compound, two albumin batches	0.19 log	1.5x	10 pairs
Within-study chain-length line, leave-one-out	0.44 log	2.8x	12 groups
Two labs, same **measured** log K	0.59 log	3.9x	46 pairs

What is being compared	Disagreement	Fold	n
Kd -> K with n assumed = 1 (Chen 2025)	0.80 log	6.3x	22 pairs
Two labs, the same albumin Kd, same method	**1.45 log**	**28x**	4 compounds

The last row is the input to the conversion, and it is the largest number in the table.

Detail. Chen 2025 and Gao 2019 both measure PFAS on human serum albumin by equilibrium dialysis - the same technique, the same protein, the same species:

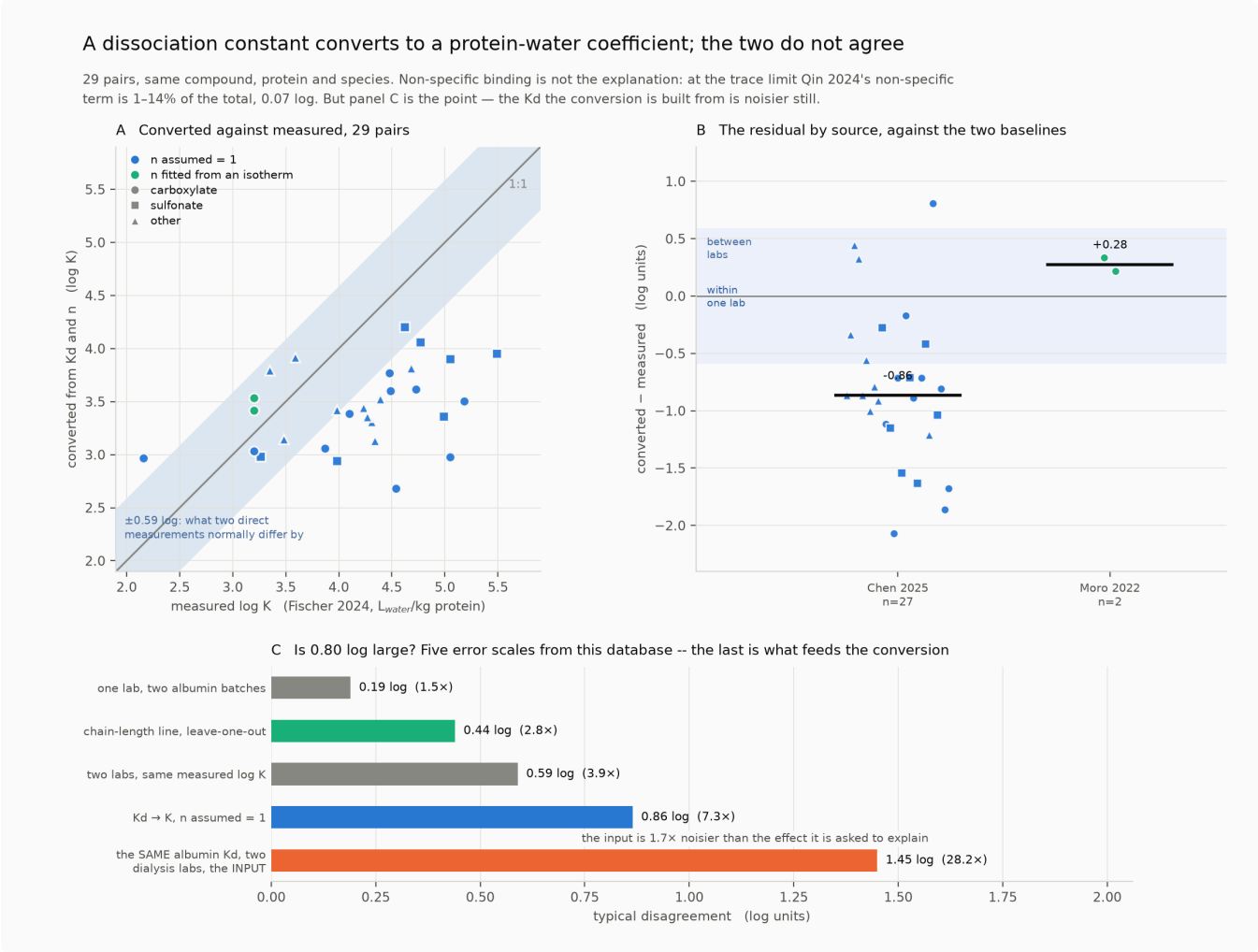
Compound	Chen 2025	Gao 2019	Gap
PFHxA	13.98 uM	195 uM	1.14 log (14x)
PFNA	3.75 uM	73 uM	1.29 log (19x)
PFOA	2.57 uM	115 uM	**1.65 log (45x)**
PFOS	0.94 uM	38 uM	1.61 log (40x)

Since `log K = log10(1e9/M) + log n - log Kd`, the converted value inherits that error **one for one**. The input is 1.8x noisier than the effect we set out to attribute. **An n wrong by six-fold is a second-order problem next to an input reproducible to 28x.**

Direction of the effect. It does *not* rescue the conversion - a one-sided offset across 22 compounds is not what noise produces. It says the offset is not worth attributing to any single cause while a larger, unattributed error sits underneath it.

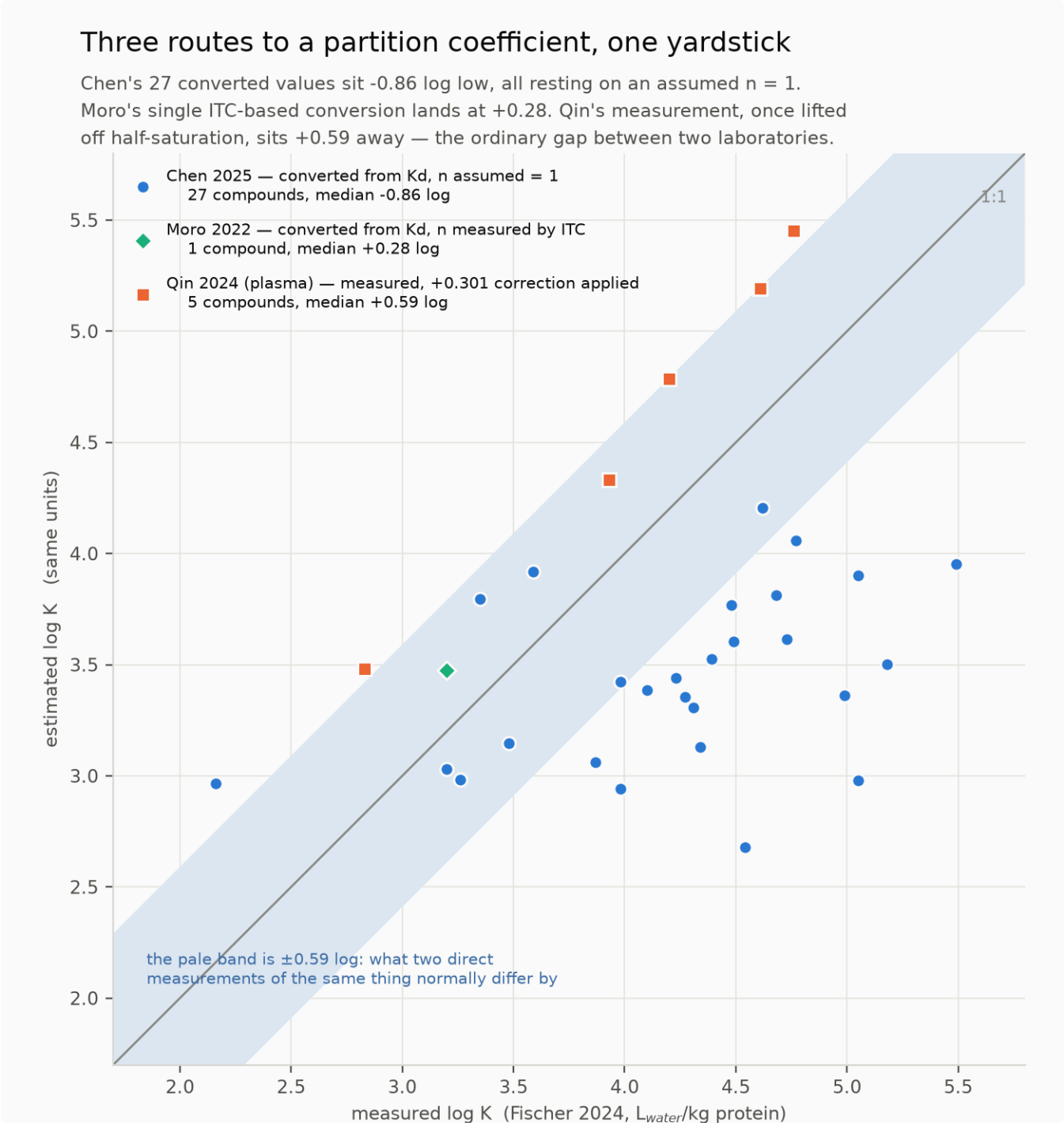
This result is independent of the conversion. PFOA on human serum albumin is the most-measured protein-ligand pair in PFAS toxicokinetics, and two equilibrium-dialysis measurements of it differ by 45x. Any model that takes one of them as *the* value inherits that, and most models take one.

Note also the asymmetry in the table: a directly measured log K is reproducible to 3.9x between labs, while the Kd is reproducible to 28x. Where a mass-balance model can be built on either, it should be built on the measured coefficient.



Panel C is the one to read first: it answers "is 0.80 log big?" by putting it beside what agreement normally looks like in this database.

The same comparison restricted to three sources. The figure above carries every admitted pair and the error scales behind them. This one answers the narrower question - does an estimate land on its measurement - for three sources chosen because each reaches a coefficient a different way, all against Fischer 2024.



Chen 2025 converts from a Kd with the stoichiometry assumed at 1 and sits 0.86 log low across 27 compounds. Moro 2022 converts the same way with n measured by calorimetry and lands at +0.28. Qin 2024 is not an estimate at all but the case that needs the correction factor: its albumin column is Eq. 15′s half-saturation ratio, so it is lifted 0.301 log onto the plateau and its specific and non-specific terms summed, after which it sits +0.59 from Fischer - the ordinary distance between two laboratories, which is the effect of the correction.

One caveat on the middle series. Moro contributes **one** compound, so it illustrates the argument rather than carrying it; the weight is in Chen's 27 against Fischer, and in the five error scales. The data is in db/actual_vs_estimated_simple.csv.

8. Affinity ranks sites, not compartment capacity

A tighter Kd on liver FABP than on albumin does **not** mean the liver holds more PFAS. The mass a phase holds at trace is

$$\text{bound per litre of phase} = C_{protein} \cdot n \cdot \frac{C_w}{K_d}$$

so a comparison between compartments is a ratio of `C x n / Kd`, and the concentration term is the one **no binding paper reports**. Until we went and got it, our database had every Kd and no C, which is why this question could be asked and not answered.

The concentrations. Human hepatic FABP1 is **399 uM of cytosol** (211-760 across 61 livers, CV 24.8%), by surrogate-peptide LC-MS/MS. Plasma albumin is 527-752 uM (the 35-50 g/L interval).

These are **the same order**. The expectation that an intracellular binding protein is too scarce to constitute a compartment does not hold for L-FABP: per litre of cytosol it is 0.63x the molarity of albumin per litre of plasma.

The comparison, with no cross-study step. Jia 2022 measures both proteins by thermophoresis in one table, so five compounds carry both constants from one run:

Compound	L-FABP Kd (uM)	HSA Kd (uM)	Capacity ratio, per L cytosol
HFPO-TeA	1.22	179	**93x**
8:2 Cl-PFESA	5.39	31	3.6x
HFPO-DA	267.8	1156	2.7x
HFPO-TA	19.94	84	2.7x
PFEESA	3138	874	**0.2x**

Geometric mean 3.4x per litre of cytosol, 1.9x per litre of liver. Four of five favour liver; the spread across five compounds in one run is **527-fold**.

The ratio factorises exactly into a flat 0.63x concentration term and a compound-specific affinity term. So the concentration we went looking for turns out not to be what decides the answer - but that could only be established by having it. Had FABP1 come back at 1 uM, the affinity term would have been irrelevant.

Conditions that would overturn this. The ratio carries `n_HSA / n_LFABP` and thermophoresis reports neither. At Maso 2021's four albumin sites against one on L-FABP, every ratio falls four-fold and **only HFPO-TeA still favours liver**. The comparison also ignores interstitial albumin in the liver, which can only push the ratio down, and all of it is the trace limit.

9. Species and sex

Two questions with sharply different answers, and the contrast is the useful part.

9.1 Species

Every row carries a species: 584 human, 170 bovine, 162 mouse, 99 rat, 31 rainbow trout, 8 porcine. Two sources compare species **within one laboratory**, which is the only comparison worth running given that two laboratories sit 1.45 log apart on the same constant.

Fischer 2024, human against bovine albumin, 27 shared compounds, one method: median difference **-0.02 log**, |median| 0.09 (1.2x), human higher in 12 of 27. That is no difference at all. The three largest are PFBS (+0.98), PFHpS (+0.61) and 5:3 FTCA (+0.58).

Starnes 2024, four species, seven compounds, one DSF run (Kd in mM):

Compound	Porcine	Rat	Human	Bovine	Spread
PFBA	1.28	1.90	2.64	3.06	2.4x
PFHxA	0.95	1.33	1.71	2.17	2.3x
PFOA	0.37	0.40	0.79	0.99	2.7x
PFBS	1.25	1.41	1.68	0.80	2.1x
PFOS	0.38	0.42	0.69	0.86	2.3x
HFPO-DA	2.12	1.64	1.57	2.55	1.6x
6:2 FTSA	0.52	0.33	0.41	1.39	4.2x

Median spread **0.36 log (2.3x)**. The order porcine < rat < human < bovine holds for 4 of 7 - an ordered effect, but a small one. 6:2 FTOH is a non-binder in all four species, which is the cleanest species agreement there is.

Caveat on those Kd values. Differential scanning fluorimetry measures binding at the protein's melting transition, and its constants run about three orders weaker than equilibrium-dialysis values for the same pairs. Because all four species were run together in one assay, the **ordering and the spread are sound**; the absolute Kd values are not physiological affinities and should not be lifted into a model.

Put against the yardsticks, both species comparisons fall below the floor:

	log	Fold
Human vs bovine albumin, one lab	0.09	1.2x
One lab, two albumin batches	0.19	1.5x
Four mammals, one lab	0.36	2.3x
Two labs, same measured log K, same species	0.59	3.9x
Two labs, same albumin Kd, same method	1.45	28x

Changing the mammal moves a PFAS-albumin constant less than changing the laboratory does. Two consequences: borrowing an albumin constant across mammals costs less than borrowing one across papers within a species - the usual worry is backwards; and a 0.36 log species effect cannot be resolved by pooling literature values, because the pooling adds 0.59 to 1.45. It needs one lab measuring both species.

9.2 Species differences in the matrix rather than the protein

Albumin is albumin. Plasma is not plasma. Qin 2024 puts human against rainbow trout **plasma** at a median -0.45 log over 16 compounds. But Qin also measures the composition - human plasma 42.25 mL/L protein and 4.46 mL/L lipid, trout 15.46 and 7.08 - and those two compositions alone confine the ratio to 0.30x-1.32x for *any* pair of coefficients. Five of eleven measured ratios fall outside that bound.

So the surviving species difference is in how much protein and lipid the fluid contains, plus something composition cannot explain - not in the protein's binding constant.

9.3 Sex

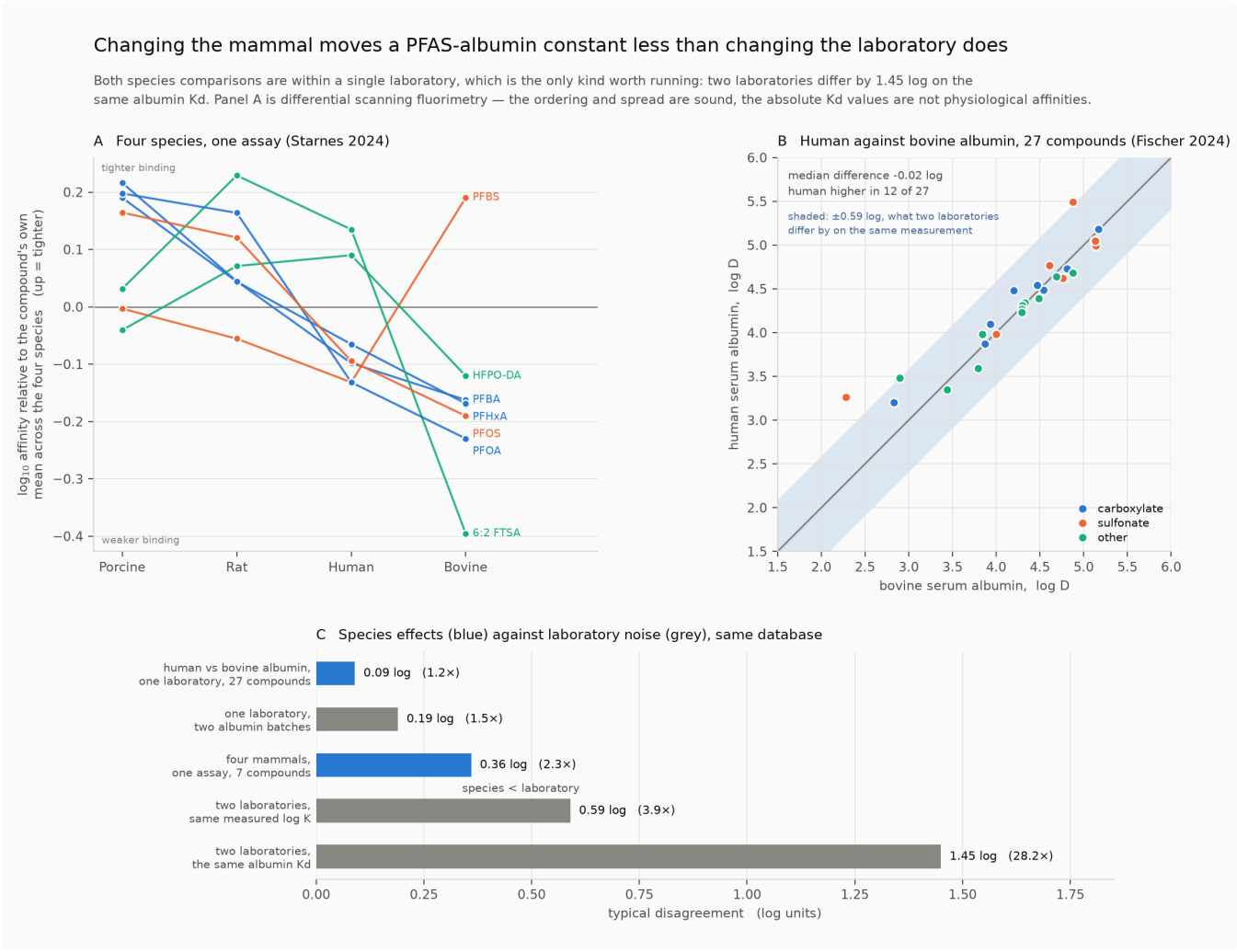
There is no sex field, and 6 of 1,241 rows mention sex anywhere in free text, from 4 studies. This is not an extraction gap. A binding assay uses purified protein, and purified albumin has no sex, so no paper reports it.

That matters because the toxicokinetic side of this project shows enormous sex effects:

PFAS	Species	Half-life F/M	Clearance F/M	% from clearance
PFOA	rat	0.049	44.2	82.4%
PFNA	rat	0.053	20.6	95.4%
PFHxS	rat	0.059	20.5	94.6%
PFBA	rat	0.281	3.6	98.3%

Rat PFHxS eliminates 17x faster in females, and 94.6% of that is **clearance** - renal transport under androgen control - not binding.

Species comparisons. Both of them, against the laboratory noise they have to be read against.



Panel A is Starnes 2024′s four mammals in one differential-scanning-fluorimetry assay, each compound plotted against its own geometric mean so that up is tighter binding. Porcine albumin binds tightest for 4 of the 7 compounds and bovine most weakly for 6 of 7, and the full ordering porcine to rat to human to bovine holds for 4 of 7; the median spread across the four species is 0.36 log and the widest, 6:2 FTSA, is 0.62.

Panel B is Fischer 2024′s 27 compounds measured on human and on bovine albumin in the same laboratory. The median difference is -0.02 log and the median absolute difference 0.09; human is higher for 12 compounds and bovine for 14, with one tie, so the sign is a coin toss rather than a species effect. Only PFBS (0.98) and PFHpS (0.61) fall outside the 0.59 log band.

Panel C sets both species scales (blue) beside the three laboratory scales (grey). Both species bars sit below the 0.59 log that two laboratories reach measuring the same quantity, and the largest bar on the chart - 1.45 log between two laboratories′ albumin

dissociation constants - is four times the largest species effect. That is the whole argument in one panel: the noise in who measured a constant is larger than the signal in which mammal it came from.

Implication for extrapolation: binding is the part of PFAS toxicokinetics that travels between species and sexes most safely. Albumin constants move less than 2.5x across four mammals and cannot move at all with sex. Transport is the part that does not travel. A model that gets species or sex extrapolation wrong is almost never wrong in its binding term.

10. Extension beyond the ten sources: differences by protein

The top-ten scope was chosen for *measured constants against a named single protein*. Extending to everything answers two questions - and the first has a dull answer.

10.1 Effect of extending the scope

Sorbent	Rows	Compounds	Studies	Share of named-protein rows
Albumin	540	192	47	**83%**
Phospholipid membrane	477	103	8	-
Whole plasma/serum	99	41	6	-
L-FABP	38	20	8	5.8%
gamma-globulin	35	29	6	5.4%
PPAR	33	13	3	5.1%
Transthyretin	2	2	2	0.3%
I-FABP	2	2	1	0.3%

Of the 650 rows against a named single protein, **83% are albumin**. The two largest blocks outside the top ten are not protein binding at all - phospholipid membrane is the *lipid* limb of a mass balance, and whole plasma is the thing a mass balance has to predict.

Convertible rows rise from 57 to 82, but 16 of the 25 added come from fluorescence double-log slopes or from a source whose affinities sit three orders below every other measurement of the same compound. What the extension genuinely buys is Khazaee 2021 and Woodcroft 2010 - the only non-albumin protein that converts at all.

The field is not short of PFAS-protein data. It is short of PFAS-protein data about any protein except albumin.

10.2 Chain-length slope, not level

Levels differ by about a log, as expected. What matters is that the **chain-length slope** differs two-fold, and albumin is the outlier. Every fit is within one paper, one method, one head group and one species (carboxylates shown; sulfonates are steeper throughout and preserve the ordering):

Sorbent	per CF2	r	n	Source	Behaves like
Albumin, specific site (-log Kd)	**+0.026**	+0.19	11	Chen 2025	a saturating site
Albumin, whole partition (log D)	+0.193	+0.64	10	Fischer 2024	a mixture
Liver FABP (-log Kd)	+0.347	+0.89	4	Jia 2022	hydrophobic partition
Phospholipid membrane (log K_MW)	+0.386	+1.00	11	Chen 2025	hydrophobic partition
gamma-globulin (log D)	+0.396	+0.89	9	Fischer 2024	hydrophobic partition

Three unrelated sorbents agree on what hydrophobic partitioning looks like: +0.35 to +0.40 per CF2. **Albumin's saturable site is flat.** Albumin's whole partition sits between the two because it *is* the two - a flat specific site plus a climbing non-specific term.

That is the same result the conversion analysis reached from the other end. The stoichiometry needed to reconcile a Kd with a partition coefficient rises with chain length; this says why. (The L-FABP fit rests on four compounds and is indicative only.)

Even the membrane series, the cleanest in the database at r = 1.00, is not a straight line: its per-CF2 steps rise from +0.39 at C4 to +0.47 at C8, then fall to +0.25 at C13. Pure lipid partitioning saturates in chain length too, just later.

10.3 Consequence: albumin's advantage declines with chain length

Fischer measured albumin and gamma-globulin on the same compounds in the same hands, so subtraction is the whole analysis - no model enters it.

Compound	nCF	Albumin	Globulin	Albumin advantage	Globulin's share of plasma protein binding
PFHxA	5	3.20	1.73	+1.47	1-3%
PFHxS	6	4.77	1.95	+2.82	<1%
PFOA	7	4.48	2.16	+2.32	<1%

Compound	nCF	Albumin	Globulin	Albumin advantage	Globulin's share of plasma protein binding
PFOS	8	4.62	3.33	+1.29	2-5%
PFDA	9	4.73	3.67	+1.06	3-8%
PFDS	10	5.05	4.96	+0.09	25-45%
FDSA	10	4.64	4.95	**-0.31**	45-67%
PFDoDA	11	5.05	4.81	+0.24	19-37%
PFTeDA	13	3.87	4.00	**-0.13**	35-57%

The last column carries only the clinical mass split (globulin 0.40-1.00x albumin, from the 35-50 and 60-80 g/L intervals) and these two measurements. Median albumin advantage: **+1.56 log at nCF <= 6, +0.12 log at nCF >= 10** - a fall of 1.44 log across the measured range, reversing outright for FDSA and PFTeDA.

Treating plasma binding as albumin-only is a chain-length-dependent error, not a constant one. For short compounds globulin is a rounding error. For long ones it supplies a third to two-thirds of the binding, and the correction biases the free fraction *high* for exactly the compounds that accumulate.

10.4 Gaps no extension closes

Transthyretin is the sharpest gap: it carries thyroxine, the displacement hypothesis for PFAS thyroid effects runs through it, and the database holds **two rows**. And no protein other than albumin has a measured protein-water partition coefficient anywhere - every non-albumin number is an affinity, which is why the conversion matters and why it cannot be checked.

11. Dataset statistics

11.1 Compounds with an albumin affinity

540 albumin rows over 192 distinct chemical entries. 489 of those name one compound; the rest are prose lists. **1**11 compounds carry a numeric affinity on albumin - 85 with a dissociation constant, 106 with an association constant, 80 with both.

Every compound count here merges spelling variants first. The same compound is written differently by different groups (PFUnA/PFUnDA, 6:2 FtS/6:2 FTSA, FOSA/PFOSA, PFECHS/PFEtCHxS), and an un-normalised count is inflated: raw names give 138 compounds with an albumin affinity where there are 111.

"Binds albumin" is doing some work in that sentence: a few are explicit **non-binders** measured and reported as such, which is a result rather than an absence. 6:2 FTOH is a non-binder in all four species Starnes tested.

11.2 Most frequently measured compounds

Compound	Rows	Studies
PFOA	38	21
PFOS	30	16
PFHxA	22	11
PFHpA	20	9
PFNA	20	12
HFPO-DA (GenX)	20	10
PFBS	19	11
PFHxS	17	10
PFBA	17	9
PFDA	14	9

Those ten carry **219 of the 465 albumin affinity rows (47%)**, and 43 of the 111 measured compounds (39%) appear exactly once. The distribution is the point: PFOA and PFOS are measured 38 and 30 times across 21 and 16 studies, and the long tail is measured once. Any statement about "PFAS binding to albumin" rests on two compounds unless it says otherwise.

11.3 Available partition coefficients

Kind	Rows	Compounds
Measured, on a stated basis	**187**	36
Biomimetic, basis never stated	56	-
Converted here from affinity + stoichiometry	82	-

Measured ones by sorbent: BSA 83 (across three batches), HSA 28, gamma-globulin 28, human plasma 21, trout plasma 16, HEK293H structural protein 11.

Coverage overlap, by compound: 36 have an affinity but no measured partition coefficient; 3 have a coefficient but no affinity; **33 have both** - the only ones where the conversion can be checked at all.

36 compounds reach a measured coefficient against 69 that reach an affinity, so the affinity literature is roughly twice the size of the partition literature on matched terms, and far more than that on the wider albumin count. Either way, that gap is why the conversion is worth having.

11.4 Where the conversion is applicable

The best agreements, converted against measured:

Compound	nCF	Converted	Measured	Residual	From
PFOA	7	4.37	4.48	-0.11	Chen & Guo 2009
PFHpA	6	4.23	4.10	+0.13	Peng 2024
PFOA	7	4.64	4.48	+0.16	Peng 2024
PFHxA	5	3.03	3.20	-0.17	Chen 2025
PFHxA	5	3.42	3.20	+0.22	Moro 2022
PFBS	4	2.98	3.26	-0.28	Chen 2025
7:3 FTCA	7	3.92	3.59	+0.33	Chen 2025
5:3 FTCA	5	3.15	3.48	-0.33	Chen 2025
PFOS	8	4.20	4.62	-0.42	Chen 2025

11 of 33 pairs agree within the 0.59 log between-laboratory floor; 4 agree within the 0.19 log within-laboratory floor.

Sorted by chain length:

Band	Pairs	Median residual	Within 0.59 log
nCF <= 6	13	-0.33	7 of 13
nCF 7-8	14	-0.90	4 of 14
nCF >= 9	6	-1.39	**0 of 6**

Every good match is nCF <= 8 (the eleven are at 4, 5, 5, 5, 5, 6, 6, 7, 7, 7, 8). Not one of the six pairs at nCF >= 9 agrees within the floor.

Rule: convert below nine perfluorinated carbons; do not convert above it.

This is not a fitted cutoff; it is what the flat specific site predicts. At short chain lengths the saturable site *is* the binding, so a single-site Kd describes the whole of it. By C9 the non-specific term has overtaken it, and a single-site Kd describes a minority of the capacity. The slope table above is the mechanism.

11.5 Compounds requiring the reverse conversion

Three compounds have a measured partition coefficient and **no affinity anywhere** in the database:

Compound	nCF	Measured in	log K	Source
10:2 FTOH	10	trout plasma / BSA / human plasma	6.72 / 6.58 / 6.08	Qin 2024
8:2 FTOH	8	trout plasma / BSA / human plasma	4.88 / 4.47 / 4.42	Qin 2024
FDSA	10	gamma-globulin / BSA / HSA	4.95 / 4.69 / 4.64	Fischer 2024

Those are exactly the compounds where the reverse conversion would earn its keep, and it is the direction that cannot be done. The only route to an affinity behind those nine measured numbers is a new saturation experiment.

Caution. Without merging spelling variants, *ten* compounds appear to have this gap. Seven are the same compound under two names, with the affinity sitting under the other spelling. A coverage statistic is only as good as its compound identity, and this one was wrong by a factor of three before the names were reconciled.

11.6 Name reconciliation

That finding prompted a reconciliation of every chemical name in every dataset, not only those in the coverage count. It lives in one module (`pfas_names.py`), covers all five source files, and is published as `db/pfas_name_map.csv`, which gives each raw name, its canonical form and the evidence.

The raw `chemical` string is never rewritten - that is provenance. Reconciliation is a layer on top, so it stays auditable and reversible. The workbooks carry a **`pfas_canonical`** column beside the source's own spelling; group and count on that one.

2**56 raw names across the five files reduce to 137 compounds, with 44** spellings merged. Three rules decided each group, in order:

1. **If one source uses both names, they are different compounds.** This separates N-TAmP-FHxSA from PFHxSaAm: Chen 2025 reports both, at 33.68 and 6.19 uM.
2. If the names share chain length and head group and no source uses both, they are candidates - merged only where the chemistry is unambiguous (an N- prefix, a trade name in brackets, a case difference, a known abbreviation pair).
3. Anything left over is **not** merged and is recorded with its reason.

What that last category holds: linear and branched PFOA/PFOS are kept apart from the technical mixture, because an isomer fraction is not the mixture. Cl-PFOS and 8Cl-PFOS are kept apart because Muensterman reports no chain length or head group, so the chlorine position cannot be confirmed. The potassium and ammonium salts are kept apart from their acids because Sakolish 2026 doses both side by side - rule 1 again. DONA is merged with ADONA, reviewed and confirmed rather than assumed. **The audit is itself audited.** A self-test blanks the alias table and asks the audit to rediscover the merges already encoded. The first version failed it - 13 of 34 known merges were invisible, because two of its generators need a chain length and head group that the sparsely annotated sources do not report. Two more generators close the gap: one on appearance (shared prefix or suffix, transposed letters, high character similarity, constrained so chain length and head group must agree where known), and one that splits a bracketed synonym, since "HFPO-DA (GenX)" is the database itself saying GenX and HFPO-DA are one compound - something no string similarity between GENX and HFPODA would ever find. Each repair surfaced merges nobody had spotted: the two reference-ligand fatty acids, PFPrS written with its systematic name, a compound whose bracketed synonym contains a comma and so had been misfiled as a prose list, and then seven more once the other three datasets came into scope, including a bare DSSTox identifier (DTXSID1037303) standing in for PFHpA, resolved from this project's own httk table rather than guessed. The self-test now passes on all 59 discoverable merges; three further merges are excluded as undiscoverable by any spelling or annotation rule and carry their evidence instead. The audit reports 47 candidate groups, every one settled by rule 1, and nothing awaiting a decision.

Applying it across the database recovered six pairs that were previously invisible, taking the converted-versus-measured comparison from 33 pairs to 39 and Chen 2025 from 22 to 27. Every conclusion above survived and the central one sharpened: the C9

boundary still holds with no exceptions, now over seven long-chain pairs instead of six. The floor agreements moved with it, from 11 and 4 on the 33 pairs the tables above were computed from to 12 and 4 on the 39.

Ten of the 39 pairs were subsequently withdrawn. Wu 2024, Peng 2024 and Chen & Guo 2009 take their stoichiometry from a fluorescence double-logarithmic or Scatchard plot, where n is a fitted slope rather than a counted site. The three scatter across -4.01 to +0.13 log, and that scatter is what establishes the slope cannot serve as n. The database now refuses the conversion on the METHOD rather than on the wording of a spreadsheet cell: Gao 2025 writes "n_Hill" and had always been rejected, while Wu writes a bare 0.681 and the identical quantity was being accepted. Peng's values of 1.1 to 1.5 look like perfectly reasonable site counts and are refused with the rest, because judging a method by whether its answer looks right is the error this exercise exists to avoid.

The cost of that exclusion is as follows. The comparison is now 27 of 29 pairs from Chen 2025. What had looked like corroboration from four other laboratories was three laboratories sharing one unreliable route to n, and the only independent converted source left is Moro 2022, with two ITC pairs. The central numbers barely move - the median goes from -0.87 to -0.81 log and the share agreeing within the between-laboratory floor stays at 31%, 9 of 29 against 12 of 39 - and the C9 boundary still holds with no exceptions, every agreeing pair at nCF 8 or shorter. But the evidential base is narrower than the earlier text implied, and the assumed-n confound is now almost total rather than merely dominant.

12. Practical rules

1. **Convert only within one paper.** It must supply the affinity and the stoichiometry together. Taking a Kd from one study and a coefficient from another gives a reconciling n spanning 752-fold, ordered by chain length - a diagnosis of why the rule exists, not a method.
2. **Carry the protein's own mass.** `log10(1e9/M)`, not 4.1774 by habit.
3. **State the basis on every output.** Per kg and per litre differ by 0.135 log. A number without its basis is unusable.
4. **Label a converted value what it is**: the specific term, not the partition coefficient. Above about C8 the specific term is the smaller of the two contributions.
5. **Convert below nine perfluorinated carbons, not above.** Every converted value that agrees with a measurement is nCF <= 8; none of the six pairs at nCF >= 9 does. This is not a fitted cutoff - it is what the flat specific site predicts.
6. **Never convert in reverse.** It is structurally impossible, not merely unextracted.

7. **Prefer a measured coefficient where one exists.** It is reproducible to 3.9x between labs; a Kd is reproducible to 28x.
8. **Where only an interpolated log K is required**, a within-study chain-length line (one study, one head group, one protein, inside the measured range) gives 0.44 log and is unbiased by construction. It cannot tell you anything about saturation, because it contains no Kd.

12.1 Conversion inventory

	Rows
Albumin rows in the database	543
Convertible forward (paper gave n)	82
Convertible in reverse	0
FABP and PPAR rows	55
...of those, convertible	5
Blocked by no stoichiometry	18
Blocked by unstated construct mass (PPARs)	7

12.2 Sources

The ten sources supplying the most measured constants against a named single protein, which is the scope of this document:

Rank	Source	Measured rows	Compounds	Quantity
1	Fischer 2024	83	29	log D - HSA, BSA, globulin
2	Chen 2025	57	57	Kd - HSA
3	Muensterman 2026	56	56	log D - HSA, biomimetic
4	Gao 2025	42	10	Ka - BSA, NaCl series

Rank	Source	Measured rows	Compounds	Quantity
5	Qin 2024 (plasma)	32	16	log D - BSA, by concentration
6	Starnes 2024	32	8	Kd - albumin, 4 species
7	Starnes 2025	28	28	Ka - HSA, DSF
8	Allendorf 2019	24	14	log D - BSA, dialysis
9	Jia 2022	13	9	Kd - hL-FABP + HSA, MST
10	Bischel 2011	11	11	log D - BSA

Findings here that rest on papers outside that set, flagged explicitly:

- **Gao 2019** (ES&T 53:6529) - the 1.45 log albumin Kd discrepancy, the single most consequential number in this document.
- **Khazaee 2021** (Toxics 9:45) - the L-FABP equilibrium-dialysis constants and the PPAR rows.
- **Woodcroft 2010** (ETC 29:1669) - the multi-site worked example.
- **Maso 2021** - four PFOA per albumin by co-crystallography.
- **Wen & Isoherranen 2026** (JPET 393:104309) - hepatic FABP1 at 399 uM, the concentration that made the capacity comparison possible.
- **Moro 2022, Peng 2024, Wu 2024, Chen & Guo 2009** - the 11 non-Chen converted pairs, 10 of them since excluded.

12.3 Reproducibility

All figures come from scripts in the repository, each of which prints its own derivation: `kd_k_interconversion.py` (the conversion and its row inventory), `logk_converted_vs_measured.py` (the 29 admissible pairs, the 10 excluded, and the five error scales), `lfabp_capacity_vs_albumin.py` (the capacity comparison), `make_logk_conversion_figure.py` (the figure), binding_species_sex_resolution.py (species and sex), binding_all_data_protein_specific.py (the full database and the protein-specific slopes), make_species_figure.py (the species figure), actual_vs_estimated_simple.py (the three-source comparison and its CSV), pfas_molar_mass.py (molar masses computed from formulae, which is what lets a concentration stated in ug/L be made commensurate with a Kd in uM),

make_formula_images.py (the twelve display formulas, which are images because the PDF and HTML exports emit a LaTeX block as code rather than typesetting it; that script holds the LaTeX source for every one).